

MineXplore: An Open-Source Reinforcement Learning Exploration Benchmark for GNSS-Denied Underground Environment

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Abstract—We present MineXplore, an Open-Source high-fidelity MuJoCo-based simulation testbed for underground robot navigation constructed from a real-world Chilean copper mine survey. Starting from raw survey data, we reconstruct a large-scale 3D tunnel network that preserves key structural properties of the original environment, including irregular tunnel topology, looped connectivity, narrow corridors, and a central rock island. The resulting environment incorporates jagged wall geometry, varying terrain friction, illumination constraints, and continuous tunnel extrusion, and achieves strong geometric consistency with the source map via pixel-to-metric calibration and topdown validation overlays. MineXplore exposes a Gymnasium-compatible navigation interface under local observations, where agents receive egocentric sensor inputs and control differential-drive motion. To validate that the environment is navigable, we train baseline reinforcement learning policies for exploration-driven navigation and measure coverage, collision rate, and episode length across the tunnel network. A representative policy achieves 92.0% coverage of the mapped free space with a 0.14 collision rate over the final 100 evaluation episodes, demonstrating that MineXplore supports reliable policy learning under realistic sensing and topology and providing a reusable benchmark for underground exploration and navigation algorithms.

Index Terms—simulation environment, underground mining robotics, reinforcement learning, MuJoCo, navigation benchmark

I. INTRODUCTION

Underground mines are among the hardest environments for autonomous ground robots. GPS is denied, lighting is degraded, dust and water reduce sensor returns, and topology is loop-rich and non-convex [3]. The DARPA Subterranean Challenge surfaced both the difficulty of such settings and the community’s best field systems [2]. Yet much of the simulation tooling used for algorithm development remains centred on Gazebo, even as GPU-accelerated simulators have become the default for modern robot learning pipelines [6], [7].

Two gaps motivate this work. First, no publicly released MuJoCo or MJX environment is grounded in a real production mine; existing subterranean simulation assets are predominantly Gazebo-based [4], [8]. Second, rare published real-mine datasets such as the Leung *et al.* 2017 Chilean underground mine dataset have been used almost exclusively for SLAM evaluation rather than as geometric sources for simulation [1], [3]. MineXplore closes this gap by compiling the Chilean survey data into a calibrated MuJoCo world that is directly

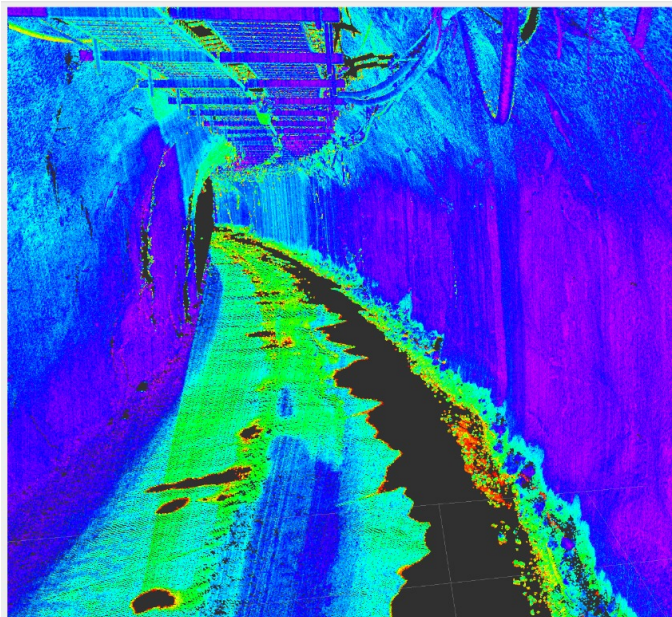


Fig. 1: Point cloud visualisation from the Chilean underground mine survey

usable for reinforcement learning under local observations. Figure ?? shows a representative point cloud from the Chilean underground mine survey, highlighting the irregular tunnel geometry and long, loop-rich corridors that MineXplore preserves in its MuJoCo environment.

We present MineXplore, a MuJoCo navigation environment whose geometry is derived from the Leung *et al.* 2017 dataset. The environment exposes a Gymnasium-style reset/step interface and supports training of reinforcement learning policies under local observations in a topologically complex layout that procedural generators such as BARN do not produce [5]. An overview of the full contour-to-MJCF compilation and validation pipeline is provided in Section III.

II. RELATED WORK

A. Underground and Subterranean Robotics

The DARPA Subterranean Challenge (2018–2021) drove recent progress in autonomous systems for GPS-denied un-

derground environments, with Team CERBERUS winning the Final Event [2], [4]. Ebadi *et al.* catalogued open problems in extreme underground SLAM across six SubT teams [3]. The Leung *et al.* 2017 Chilean underground mine dataset, our geometric source, remains the only publicly released production-mine sensor log suitable for navigation and SLAM research [1].

B. Simulation Environments for Robot Learning

Gazebo-based assets dominate the subterranean simulation niche: the SubT Virtual Testbed ships replica competition courses [4], and BARN provides 300 procedurally generated navigation worlds used in ICRA benchmarks [5]. For large-scale reinforcement learning, Isaac Gym [6] and MuJoCo Playground [7] have become standard. None of these provide a MuJoCo asset grounded in real mine survey geometry.

C. Prior Use of the Chilean Mine Dataset

The Leung *et al.* dataset provides 3D LiDAR, 2D radar, and stereo logs from a production-active tunnel approximately two kilometres long [1]. Prior work has used it exclusively for SLAM benchmarking [3]. To the best of our knowledge, no prior work has compiled its survey geometry into a physics-based simulation environment. MineXplore closes this gap.

D. Real-Mine-Grounded Simulation

Closest to our work, Gao and Awuah-Offei built a ROS 2 and Gazebo framework for quadruped navigation in 3D maps of real sites including the Edgar Mine teaching facility [8]. MineXplore differs in that (i) it targets the MuJoCo/MJX ecosystem, (ii) its source is a real operational copper mine [1] rather than a teaching facility, and (iii) it exposes a Gymnasium-compatible interface for single- and multi-agent reinforcement learning under local observations.

III. ENVIRONMENT DESIGN

Section III walks through the MineXplore compilation pipeline in order: source inputs, 2D geometry extraction, 3D compilation, visual feature mapping, and validation. Stage labels correspond to Fig. 2.

A. Source Data and Scale Calibration (Stage 1)

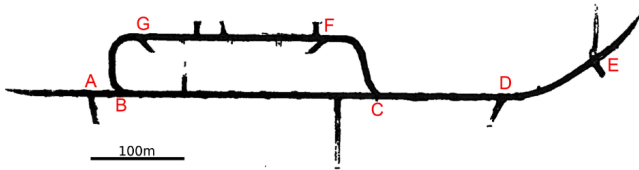


Fig. 3: Published 2D survey floor-plan of the Chilean underground copper mine

MineXplore uses two complementary inputs from the Leung *et al.* 2017 Chilean underground mine dataset [1]. The tunnel geometry is derived from the published 2D survey floor-plan

image, shown in Fig. 3, which we use directly as the geometric source. The *visual surface features* — wall roughness, jagged edges, and texture — are derived from the accompanying 3D LiDAR point cloud logs. We calibrated the pixel-to-metric scale against the 100 m scale bar on the source image, yielding 1.36 px/m. The constant tunnel height is set to 3.5 m, consistent with the production tunnel specification in the source dataset. Floor elevation changes and cross-sectional variation are not modelled; the geometry is planar and constant-height by construction (Section VI).

B. 2D Geometry Extraction (Stages 2–3)

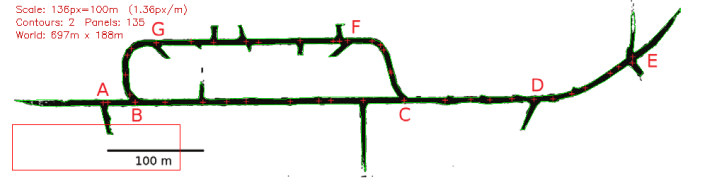


Fig. 4: Detected contours overlaid on the binarised survey map

We binarised the survey floor-plan and extracted the tunnel footprint using OpenCV contour detection. The largest external contour defines the outer tunnel boundary; interior contours define rock islands and pillars. Fig. 4 shows the detected contours overlaid on the binarised map, with the outer tunnel boundary in green and the interior rock island in red. Each contour was converted to a Shapely polygon with holes and simplified using Douglas–Peucker with a tolerance tuned to preserve corners under 1 m. The natural irregularity retained by this tolerance is the primary source of jagged wall geometry in the compiled model. The compiled footprint contains one outer boundary and one dominant interior rock island of 14,577 m², yielding a navigable area of 104,423 m² within a world extent of approximately 697 m × 188 m.

C. 3D Compilation and Physics Setup (Stages 3–6)

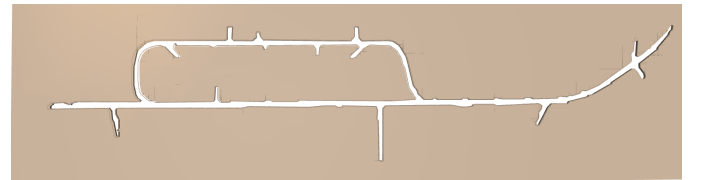


Fig. 5: Compiled MineXplore environment in the MuJoCo interactive viewer

We triangulated the 2D polygon set and extruded each face to 3.5 m (Stage 4). MuJoCo’s collision engine requires convex geoms; we decomposed the extruded mesh using V-HACD with a resolution of 1,000,000, a concavity threshold of 0.5, and a maximum of 32 convex hulls per rock tile. Applied across 70 rock geometry tiles, this produces 1,186 convex collision geoms in the final MJCF model. The world is centred at (355.3, 94.0, 1.75) m and is loaded from `chilean_mine_geometry.xml`. Ceiling and floor

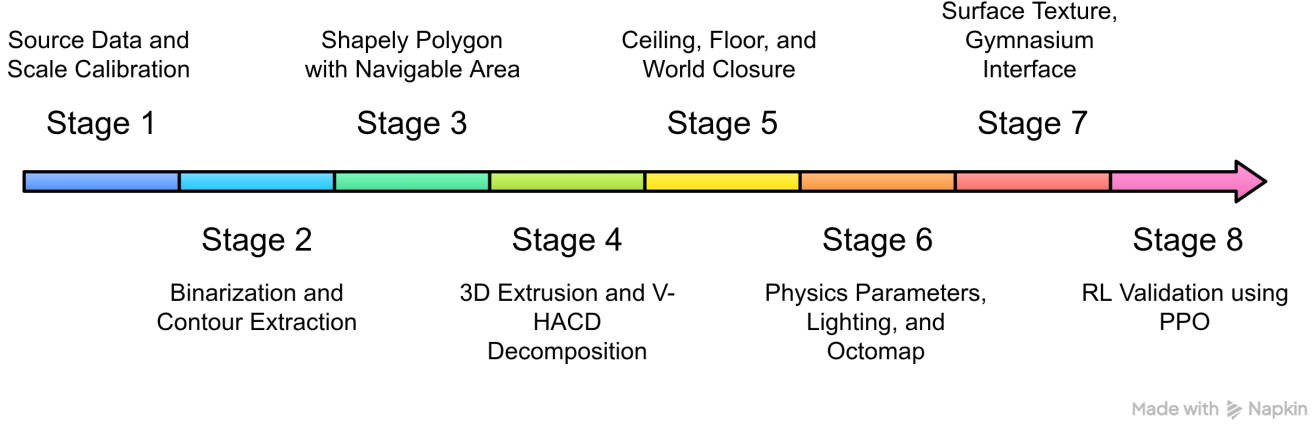


Fig. 2: MineXplore compilation pipeline: from real Chilean mine survey data through seven stages of geometry extraction, 3D compilation, surface feature mapping, and RL validation to a Gymnasium-compatible MuJoCo benchmark environment.

planes (Stage 5) close the tunnel volume. Physics parameters (Stage 6) set wall and floor friction, add directional lighting consistent with low-illumination mine conditions, and include an occupancy octomap for collision-free spawn sampling. Compilation and scene load complete in under ten seconds on an AMD Ryzen 5 5600H under Ubuntu Linux. Fig. 5 shows the compiled MuJoCo world.

D. Surface Features and Texture Mapping (Stage 7)

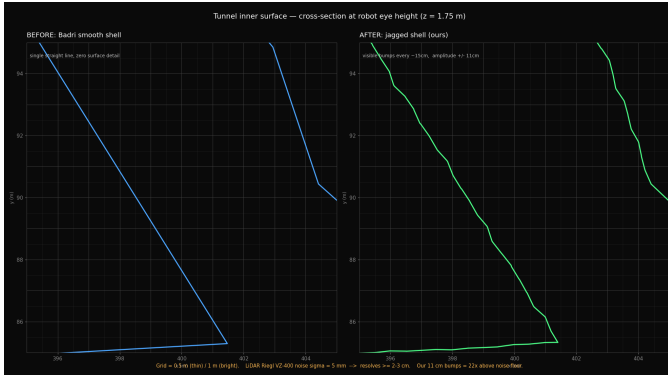


Fig. 6: Smooth flat-extruded wall (left) versus point-cloud-derived jagged surface (right) in MineXplore

Wall roughness and surface detail are derived from the Leung *et al.* 3D LiDAR point cloud logs, which capture the irregular rock surfaces of the production tunnel. We extract per-surface normal variation and roughness profiles from the point cloud and apply them as displacement and texture maps onto the compiled MJCF geometry, producing visual conditions closer to the real mine than a flat-extruded model alone would provide. Fig. III illustrates the difference between a smooth flat-extruded wall and the point-cloud-derived jagged

surface. Texture mapping is applied at the mesh level and does not affect collision geometry or physics parameters.

E. Geometry Validation

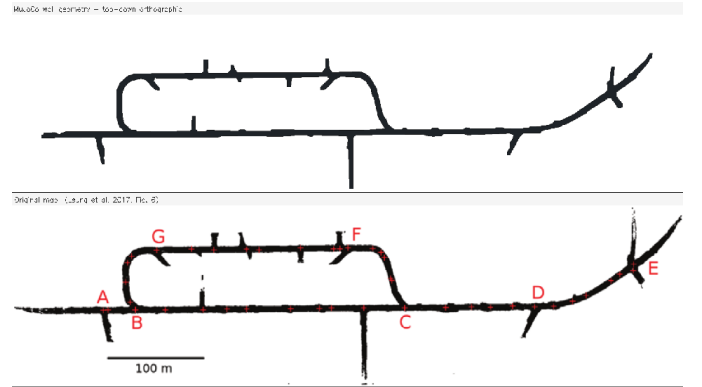


Fig. 7: Topdown overlay of the compiled MineXplore MJCF world against the original Leung *et al.* survey map

IV. BENCHMARK INTERFACE

We validated the compiled model on two axes. For *navigability*, we cast rays through the free volume and confirmed that every pixel marked navigable in the source floor-plan maps to an unobstructed column in the MJCF model. For *geometric consistency*, we rendered an orthographic topdown view of the compiled model and overlaid it on the original floor plan. Fig. 7 shows the result. The outer tunnel boundary, central rock island, and connecting passages coincide within the Douglas-Peucker simplification tolerance. This topdown overlay is the primary geometric fidelity check of the paper and is produced directly from the compiled MJCF, not from any intermediate representation.

A. Observation Space

Each agent receives a 19-dimensional observation vector at every control step. The first 16 elements are LiDAR range readings from a simulated 2D scanner with 16 uniformly spaced beams covering the full 360° field of view ($[0, 2\pi)$). Raw ranges are clipped to $[0.12 \text{ m}, 10.0 \text{ m}]$, then normalised to $[0, 1]$. Gaussian noise with $\sigma = 0.01 \text{ m}$ is applied to each beam before normalisation. The remaining 3 elements are: the previous normalised linear command, a fixed scalar of 0.0, and the previous normalised angular command, providing the agent with one step of action history. No goal-pose features and no collision flag appear in the observation vector; collision information is available only in the `info` dictionary.

B. Action Space

The action space is a continuous 2D box $\mathcal{A} = [-1, 1]^2$. The policy output (a_0, a_1) is mapped to differential-drive commands as:

$$v = a_0 \cdot 1.5 \text{ m/s}, \quad \omega = a_1 \cdot 1.5 \text{ rad/s} \quad (1)$$

Commands are then scaled and perturbed by terrain type before execution: dry terrain applies scale 1.0 with zero noise; muddy terrain applies scale 0.7 with additive noise $\sigma = 0.04$; watery terrain applies scale 0.4 with noise $\sigma = 0.10$. This terrain-dependent actuation introduces realistic motion uncertainty without altering the policy output range.

C. Episode Structure

MineXplore is a pure exploration environment. There is no goal position: `terminated` is always `False` and episodes end only by truncation. The maximum episode horizon is 6,000 control steps at a timestep of 0.1 s, giving a nominal maximum duration of 600 s. An episode is also truncated early if the agent’s speed remains below 0.01 m/s for 100 consecutive steps, preventing degenerate stationary policies. At the start of each episode, the agent is spawned at a fixed anchor near the mine entrance at $(82.0, 92.0) \text{ m}$; if that cell fails clearance checks, the agent falls back to a random valid point in Region A with uniform yaw sampling over $[-\pi, \pi)$.

D. Reward Structure

The per-step reward r is:

$$r = \Delta c \cdot g - \mathbb{I}[\Delta c = 0] \cdot 0.002 + M - \mathbb{I}[\text{collision}] \cdot 15.0 \quad (2)$$

where Δc is the number of newly discovered free-space cells at the current step, and $g = 10.0/C_{\text{total}}$ is a per-cell gain normalised by the total number of tunnel cells C_{total} . A small step penalty of -0.002 is applied on steps where no new cells are discovered to discourage stationary behaviour. One-time milestone bonuses M are awarded at 25% coverage (+15), 50% coverage (+30), and 75% coverage (+45). A collision penalty of -15.0 is applied on every step in which the agent contacts a non-floor geometry. We use reinforcement learning solely to validate that MineXplore supports stable policy learning and meaningful exploration under local observations; RL is not the primary contribution of this paper.

E. Reproducibility

In the experiments reported here we use a single-agent configuration. When multiple agents are active, each agent involved in an inter-agent collision receives the full -15.0 collision penalty independently; uninvolved agents are unaffected. All experiments use MuJoCo 3.2.6 with fixed random seeds and the compiled MJCF model under version control. Code and Environment pipeline will be made Open-sourced

V. RESULTS

A. Environment Validation

The topdown overlay of Fig. 7 confirms that the compiled MJCF model preserves the topology and scale of the source survey. Ray-cast navigability passes for 100% of sampled free-space points, confirming that no navigable pixel in the source floor-plan is occluded by collision geometry in the compiled model. Load times, geom counts, and world extents match the values reported in Section III.

B. Reinforcement Learning Baseline

We use a single-agent PPO policy to validate that MineXplore supports stable exploration learning under local observations. RL is not a contribution of this paper; the baseline exists solely to confirm that the environment is navigable and that reward signals are well-formed. Training ran for 254 episodes on a single agent with the reward structure defined in Section IV. All metrics below are reported as means over the final 100 episodes.

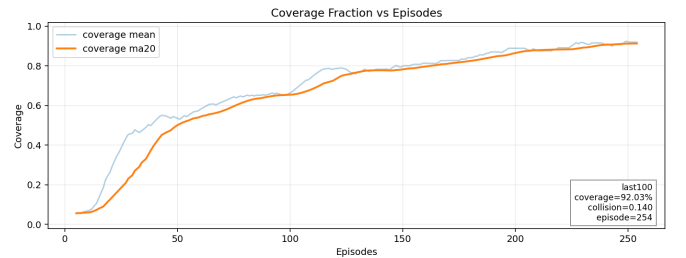


Fig. 8: Coverage fraction over training.

1) *Coverage*: Fig. 8 shows the fraction of total navigable tunnel area discovered per episode over training. Coverage rises from approximately 5% at episode 1 to **92.03%** in the final 100 episodes. The smooth, monotonic increase in the moving average (ma20) with no plateau before episode 150 confirms that the environment presents a non-trivial exploration challenge that rewards continued learning rather than saturating early.

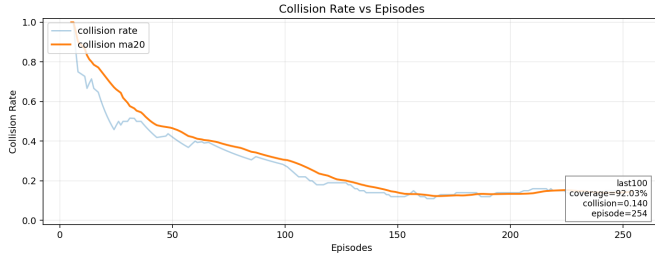


Fig. 9: Collision rate over training

2) *Collision Rate*: Fig. 9 shows the per-episode collision rate over training. The rate begins near 1.0 in early episodes, reflecting an untrained policy that contacts walls frequently in the narrow, irregular corridors. It falls steadily to **0.140** in the final 100 episodes, demonstrating that the agent learns to navigate within the V-HACD collision geometry without systematic failures.



Fig. 10: Cumulative episode reward over training

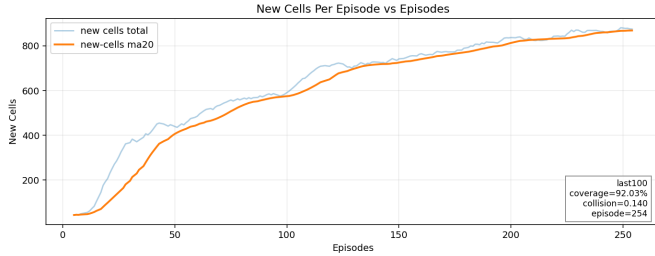


Fig. 11: New cells discovered per episode over training

3) *Cumulative Reward and New Cells*: Figs. 10 and 11 show cumulative episode reward and new cells discovered per episode respectively. Both metrics rise monotonically from negative reward (≈ -300) and ≈ 75 new cells per episode at the start, to $\approx 2,050$ cumulative reward and ≈ 860 new cells by episode 254. The alignment between these two curves directly reflects the reward structure: new-cell discovery is the primary reward signal, so reward growth is structurally coupled to exploration breadth. This coupling confirms that the reward function is well-calibrated and that no degenerate reward exploitation occurs.

Taken together, these four curves validate three properties of MineXplore as a benchmark environment: (i) the tunnel geometry is physically coherent and navigable under V-HACD

collision decomposition; (ii) the reward structure produces stable, monotonic learning signals without saturation or exploitation; and (iii) the local 16-beam LiDAR observation is sufficient for an agent to explore 92% of a 104,423 m² real-mine-derived tunnel network within 254 training episodes.

VI. DISCUSSION

A. What MineXplore Is and Is Not

MineXplore is a single-mine MuJoCo/MJX navigation environment grounded in a real production tunnel, targeted at researchers who need a non-procedural, topology-rich underground test case for navigation and exploration algorithm development. It is not a physics-calibrated digital twin: wall friction, surface roughness, and contact stiffness are MuJoCo defaults, not measurements from the source dataset. It is not a multi-mine suite, and it makes no claim of sim-to-real transfer.

B. Limitations

Two limitations bound this work. First, the collision model is a V-HACD convex decomposition with resolution 1,000,000 and concavity threshold 0.5; this may introduce localised hull artefacts in the narrowest tunnel passages. Third, the baseline results reported here use a single-agent PPO policy; broader multi-agent and cross-algorithm baselines are deferred to future work.

C. When to Use MineXplore

MineXplore is the right tool when you need a large-scale, real-mine-derived obstacle topology inside a MuJoCo or MJX training pipeline, a Gymnasium-compatible interface for single- or multi-agent navigation under local observations, or matched source data for future sim-to-real work grounded in the Leung *et al.* Chilean mine dataset [1].

VII. CONCLUSION

We presented MineXplore, an Open Source MuJoCo navigation environment derived from the Leung *et al.* 2017 Chilean underground copper mine survey. We described the contour-to-MJCF compilation pipeline (Fig. 2), validated the compiled geometry against the source survey via a topdown overlay (Fig. 7), and confirmed navigability using a single-agent PPO baseline that achieves 92.03% tunnel coverage with a 0.140 collision rate over the final 100 training episodes. MineXplore closes a specific gap in the robot-learning tooling landscape: a real-mine-grounded, Gymnasium-compatible navigation environment in the MuJoCo ecosystem. Future work includes elevation-aware geometry derived from the Leung *et al.* 3D LiDAR logs, LiDAR-replay validation against the original dataset sweeps, procedural cross-section perturbation for environment diversity, and broader multi-agent and cross-algorithm baselines.

ACKNOWLEDGMENT

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